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The Editors-in-Chief
Machine Learning
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Dear Editors,

Please find enclosed the revised version of "*VOUCH: Anytime-Valid Certificates for Machine Unlearning via Paired-Canary Hypothesis-Betting*," together with a point-by-point response to both reviewers and a tracked-changes copy of the manuscript. In keeping with double-blind review, the manuscript itself carries no author information; I am the sole author, and my identity and affiliation appear only in this letter.

Both reports were exceptionally careful, and two of their points were mathematically decisive rather than editorial. I want to flag them here because they changed the paper rather than its wording.

Reviewer 1 observed that the certificate's null was defined as a *marginal* sign probability while the supermartingale proof in fact requires a *conditional* statement at every step—and that the two come apart precisely in our setting, because every canary passes through one jointly trained and unlearned model. This is correct. Rather than patch the definition, the revision retargets the certificate at the *realised advantage on the committed cohort* and bets against a without-replacement boundary recentred at each step. The resulting guarantee is exact under arbitrary dependence across pairs, templates and repetition strata, with no independence and no exchangeability assumed. A new simulation shows the concern was not academic: under model-level over-dispersion the previous construction breaches its error level eightfold, while the corrected one holds everywhere. The same reviewer also corrected the small-ε Kullback–Leibler expansion, which was wrong by a factor of four in the stated rule of thumb, though the numerical tables had always been computed from the exact divergence.

Reviewer 2 pressed on four things the paper had asserted rather than shown, and in one case the measurement contradicted us. We had claimed the planted canaries "do not stand out." They do: a perplexity filter costing one forward pass per record recovers the entire cohort. The revision retracts the claim, explains the information-theoretic reason a compact high-entropy secret must be conspicuous, derives an entropy-dilution design rule from it, and shows the rule works (detection recall falls from 1.00 to 0.21)—while making clear why locating the cohort is not the same as defeating the audit, since both twins in a pair are equally conspicuous. Reviewer 2 also identified that the sample-size table's starred entries sat below the information-theoretic bound because they were medians over uncensored runs; the corrected Kaplan–Meier medians, with censoring rates reported, remove the artefact.

Beyond those, the revision adds four measurement studies that the reviewers asked for and that the paper is stronger for: a comparison against four *valid* sequential procedures rather than a straw fixed-sample baseline; a head-to-head against the descriptive metrics we claim to replace, on identical models and seeds; attacks from *outside* the declared score class, one of which exceeds the certified tolerance on 3.2% of certified runs and which we report as the negative result it is; and a tolerance sweep that reaches $\varepsilon = 0.05$ on real models, with the cohort versus super-population distinction made explicit rather than elided. SimNPO is added as a subject, and the benchmarks' own forget/retain metrics plus a general-knowledge probe now accompany every verdict. The scope of the certificate—a declared score class, on a committed cohort, under an honest-but-verifiable provider—is now stated consistently in the abstract, introduction, experiments and conclusion, with the measured cost of each qualification. The response letter closes with an explicit list of the three requests this revision answers only partially.

As stated in the manuscript's data availability statement, all code, per-seed results, and the execution harness are publicly released; the repository reference is withheld from the blinded manuscript and is:

`github.com/vinhqdang/VOUCH-Verifiable-Online-Unlearning-Certification-via-Hypothesis-betting`

This manuscript is original, has not been published previously, and is not under consideration elsewhere. I have no conflicts of interest to declare. I am grateful to both reviewers for reports that materially improved the work, and I hope the revision is now suitable for publication in *Machine Learning*.

Sincerely,

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